EBU6501 Alternative Assessment (Odd Semester - Paper A)

Joint Programme Assessments 2022/23

EBU6501 Middleware

Answering this paper requires **2 hours**; Answers to be submitted within the allocated **3 hours window**.

Answer ALL questions

INSTRUCTIONS

1. **You must NOT share any content from this document during the assessment period.**
2. Your answers must be typed, and diagrams or equations must be written clearly and legibly with black or blue colour **and in English**.
3. You need to submit your answers BEFORE the allocated deadline.
4. **Read the instructions on the inside cover of the questions sheet.**

Examiners

Dr Gokop Goteng, Dr Ethan Lau

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Filename: EBU6501_ALT-2022-23_S1A

Instructions

This is an open-book assessment, which should be completed **within 2 hours**. You MUST submit your answers within the allocated time in the timetable.

You MUST complete the assessment on your own, without consulting any other person. You MAY NOT check your answers with any other person.

You can refer to textbooks, notes and online materials to facilitate your working, if you provide a direct quote, or copy a diagram or chart, you must cite the source.

Before you start the assessment

1) Read the questions thoroughly and understand them.

2) Ensure you have all the resources you require to complete and upload the final assessment.

3) If you require any assistance, **raise the issue via the messaging section of this assessment on QMPlus**, immediately.

During the assessment session

1) Use the supplied answer sheet document to enter your answers. Start on a new page for each question. Make sure it is clear which question number you are answering.

2) **Type your answers** in the supplied answer sheet; hand-written equations or sketches can be incorporated into the answer sheet. Please save your work at least every 15 minutes so that you do not risk losing it.

3) When completed answering all questions, perform a word count and list the number of words on the answer sheet, then save the file as pdf before uploading, **only pdf will be accepted**, any other file format will not be accepted.

4) Your submission must be your own work, and you must ensure that you do not break any of the rules in the Academic Misconduct Policy.

Submitting the Assessment

1) You must submit your answers within the scheduled assessment time – do not leave submissions too close to the deadline. NO late submission will be accepted, no exceptions.

2) Make sure you upload and submit the final version before the deadline.

3) Please be aware that submissions will be subject to review, including but not limited to plagiarism detection software.

If you have any problems relating to access or submitting during the assessment period, please contact the email (it-issues@qmbupt.org), state the module code in the subject, and clearly state your name and student ID and any issues you are experiencing. You must use either @qmul.ac.uk or @bupt.edu.cn email address. Requests from external email addresses will not be processed.

Question 1

a) Maintaining the order, explain how the communication of messages within the Message-oriented middleware (MOM) is achieved within the objects **`MessageProducer`** and **`MessageConsumer`**. Draw the MOM architecture diagram to aid the explanation.

(5 marks)

b) Apache ActiveMQ is an example of the Message Broker for the Java Message Service (JMS) application. It is used in publishing messages to a ***Topic*** or ***Queue***.

(8 marks)

i) The Figure 1(a) and 1(b) show the type of the JMS messaging model. Denote the type of the messaging model used in a) and b). Then, explain the difference between the models.

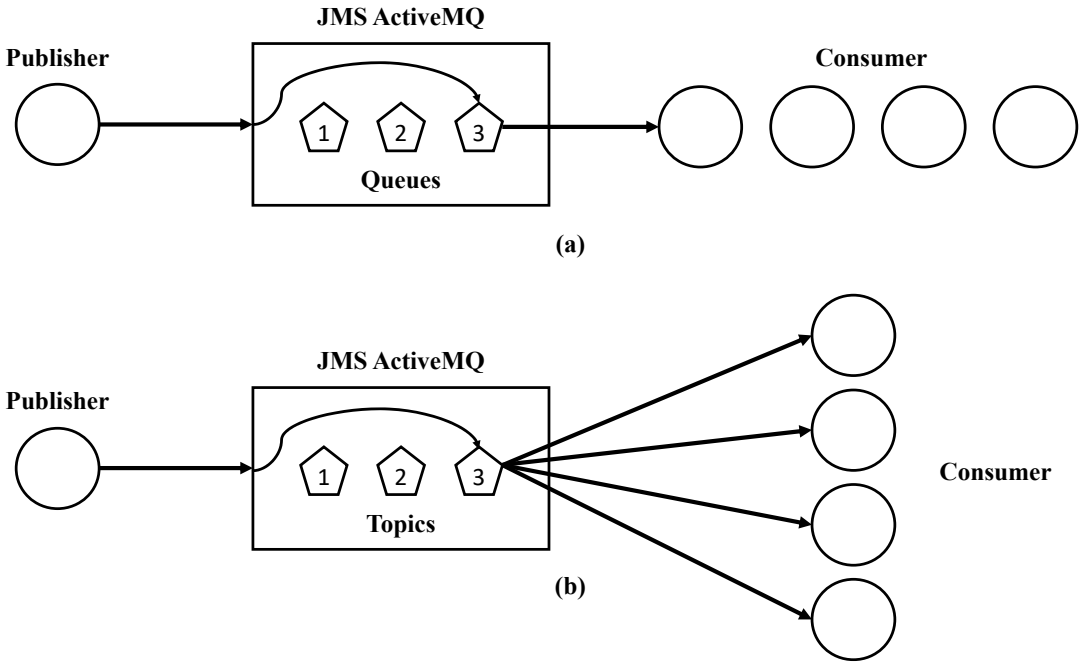


Figure 1 (a-b).

[5 marks]

ii) In your own words, describe the importance of the ***Sessions*** in the context of JMS Application Programming Interface (API) model.

[3 marks]

c) Figure 2 below shows the directory structure of the Tomcat container.

(5 marks)

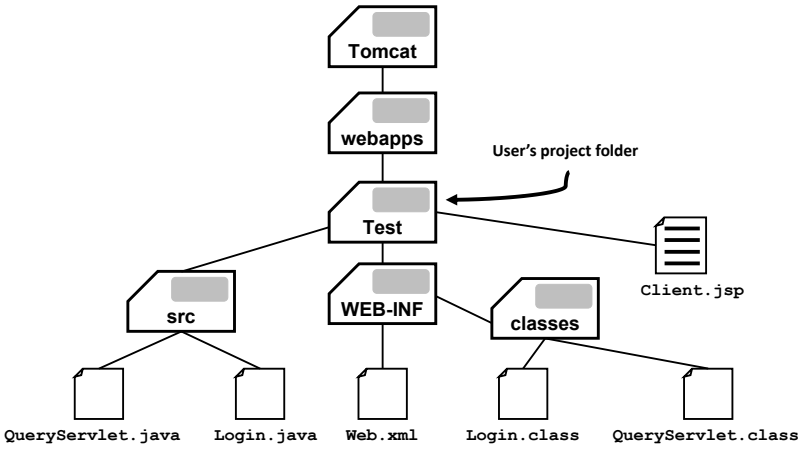


Figure 2.

i) Carefully denote the pages that are belong to the individual design pattern of the ***Model***, ***View*** and ***Controller*** model.

[3 marks]

ii) Define the term given to the file "**web.xml**" and explain why it is located under the **WEB-INF** directory**.**

[2 marks]

d) Figure 3 presents the usage of Java Servlet implementation of the ***Request-Response*** programming model which responses to client's request by querying the database. The **`QueryServlet`** is created that extends **`HttpServlet`** class. The code requires the completion of the **`Service`** and **`Destroy`** methods. The servlet shall return an HTML page that display the query response the string field **`myStatus = "<Query Response>"`**. Complete the following:

(7 marks)

```
public class QueryServlet extends HttpServlet {

      // ### Missing the service() method

      // ### Missing the destroy() method
                        }
```

Figure 3.

i) Write the required ***`service()`*** method that called by the server to allow a servlet to handle a GET request.

[2 marks]

ii) Set the content type as "text/html" for the webpage.

[1 mark]

iii) Write message ***`myStatus`*** to the webpage.

[2 marks]

iv) Close the stream for garbage collection and destroy the servlet.

[2 marks]

Question 2

a) The Inversion of Control (IoC) is a principle in the software engineering of the Spring Framework as shown in Figure 4. It is most often used in the context of object-oriented programming such as Java.

[10 marks]

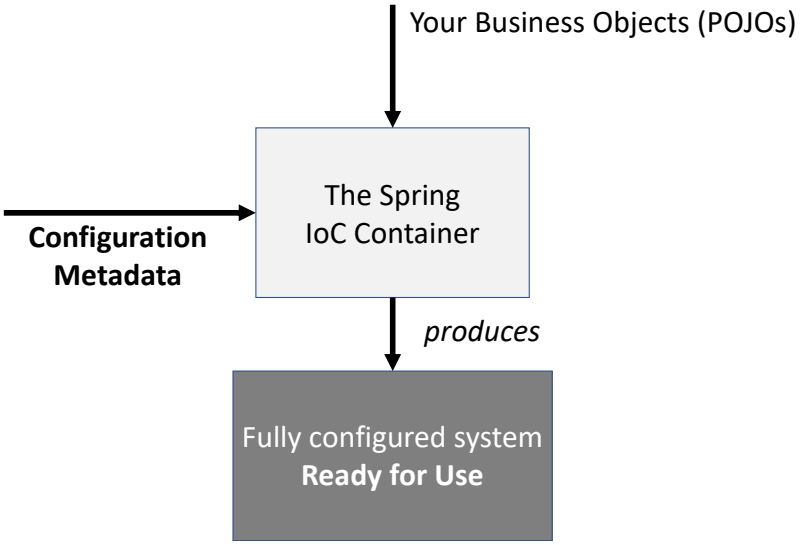


Figure 4

i) Based on the information in Figure 4, provide TWO types of the spring IoC container and further explain its role in the Spring Framework.

(4 marks)

ii) List out the TWO examples of the configuration metadata.

(2 marks)

iii) The Application Context is Spring's advanced container. Implement the Spring's Application Context by instantiating the **`ApplicationContext`** interface that loads the .xml configurations from a classpath of "**info.xml**". Finally, obtain the bean with the bean ID = **`beanInfo`** from the **`beanInfo`** class.

(4 marks)

b) This question is about Java threads.

[15 marks]

i) Maintaining their order, state and explain the main FIVE life cycle states of a Java thread.

[6 marks]

ii) In your own words, explain the ***Visibility*** and ***Race Conditions*** concurrency issues.

[4 marks]

iii) Suggest the corresponding methods to avoid the ***Visibility*** and ***Race Conditions*** issues.

[2 marks]

iv) Figure 5 shows the Java class **`MyAccumulator`** where the method **`getNextValue()`** is not thread-safe as it is accessible by multiple threads. Improve the code to ensure the thread-safe and atomic operation.

```
public class MyAccumulator {
  private int value = 500;
  public int getNextValue(){
        return value++;
  }
}
```

Figure 5.

[3 marks]

Question 3

a) An Internet of Things (IoT) company is using thousands of sensors to receive, process and send data in form of messages to heterogeneous distributed computer platforms across many countries as logs. The data come from many sources to forecast weather, mortgage rates, inflation, currency exchange rates and many others. The system that stores this data as logs is not well organised and it is difficult to identify and separate the data into different areas (weather, mortgage, inflation, etc) to make any meaning from them. A Middleware Developer is hired by the company to develop a system that will ensure data is well organised and durable using Apache Kafka.

[18 marks]

i) Describe how the Middleware Developer will ensure that the different data from the different sources will be well organised (weather, mortgage rates, etc should be separate) and durable (last for a long time) using Kafka.

(8 marks)

ii) Describe the TWO mechanisms that the Middleware Developer can use to manage and convert the internal data (machine readable) to external data (human readable) formats when data is transported from sensors to output units for users to use and vice versa.

(2 marks)

iii) As a Middleware Developer, write an Apache Kafka Consumer Library for a mortgage rate payment using Java programming language. The program will use the Java class KafkaConsumer as a Java Collection that has two parameters String and Payment (Payment is a user defined class). The program will also use the Java Collection ConsumerRecord that also use the same parameters (String and Payment).

(8 marks)

b) Describe TWO methods in which you can use **Cryptography** to ensure secure sharing of files using one key and two keys respectively in a computer network.

[5 marks]

c) Describe ONE IoT protocol that is the most commonly used in homes and is suitable for short and medium range distance coverage in a Local Area Network (LAN).

[2 marks]

Question 4

a) Using your knowledge of the Open Services Gateway Initiatives (OSGi) architecture, describe how ***Bundles***, ***Security*** and ***Services*** work in OSGi.

[6 marks]

b) A company has been using physical machines to run its software and hardware middleware systems. The company wants to move all its IT infrastructure to the cloud and is looking for a way of eliminating the time and effort of provisioning compute and storage servers. It also wants to adopt the methodology of running and processing data in automated small pieces of actions that are faster and more efficient. Describe TWO types of services that can achieve these requirements for the company using TWO examples (ONE for each service) of such services.

[10 marks]

c) Describe the FIVE main steps in using a Uniform Resource Locator (URL) in a networking Java program (use Java classes and methods where possible).

[9 marks]

END OF PAPER

DO NOT WRITE ON THIS PAGE.